

ULCINJ, Montenegro

WMO: 134640

Lat: 41.917N

Lon: 19.217E

Elev: 95

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 90-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.4	32.9	9.8	9.1	36.7	13.5	10.9	37.1	30.3	34.8	25.9	39.5	9.4	70	0.335

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	91.1	73.9	88.1	73.1	85.6	72.4	77.6	85.1	76.1	83.6	74.8	82.3	7.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	133.5	81.8	73.9	126.8	81.2	72.3	120.1	80.1	41.1	85.7	39.6	83.6	38.3	82.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.7	15.1		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.1	45.8	46.5	52.5	57.9	65.7	72.3	77.6	77.1	70.4	64.5	55.6	46.8
	DBStd	12.51	5.82	6.48	5.53	4.57	4.89	6.02	4.03	4.27	3.94	5.58	6.29	5.75
	HDD50	479	152	129	39	4	0	0	0	0	0	0	21	134
	HDD65	2711	596	518	386	217	52	10	0	0	5	77	286	564
	CDD50	4543	21	31	118	240	486	669	854	841	612	449	188	34
	CDD65	1300	0	0	0	3	72	230	389	376	167	60	3	0
	CDH74	9369	0	0	0	5	314	1726	3409	3103	673	138	1	0
	CDH80	2658	0	0	0	0	48	515	1080	911	94	10	0	0
Wind	WSAvg	5.5	6.3	7.1	5.7	4.9	4.6	4.4	4.4	4.4	4.9	5.7	5.6	7.7
Precipitation	PrecAvg	55.2	7.1	5.7	5.1	4.9	2.7	2.2	1.3	2.0	3.9	6.0	7.5	6.9
	PrecMax	70.5	12.9	13.7	10.6	10.6	6.9	5.9	5.1	9.1	10.0	15.5	14.5	17.9
	PrecMin	39.9	0.0	1.4	1.1	1.0	0.4	0.2	0.0	0.0	0.3	0.0	2.4	2.0
	PrecStd	9.3	3.5	2.9	2.8	2.4	2.2	1.7	1.2	2.2	2.8	4.3	4.1	3.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.0	63.5	68.3	74.3	84.4	92.8	94.4	93.7	86.3	81.0	72.6	64.0
		MCWB	53.3	51.9	53.5	60.2	68.6	74.4	73.8	75.9	72.0	65.7	62.7	60.1
	2%	DB	58.5	60.2	64.9	70.9	80.3	88.6	91.3	90.4	82.2	77.6	69.4	60.1
		MCWB	52.8	51.5	54.1	58.8	66.0	74.0	72.9	74.2	69.2	66.9	62.3	54.7
	5%	DB	56.6	57.8	62.6	68.0	77.4	85.4	88.5	87.1	79.8	75.1	67.1	57.7
		MCWB	51.4	50.5	53.7	57.9	65.2	72.3	72.6	73.5	68.6	66.6	60.1	53.1
	10%	DB	54.5	56.0	60.3	65.3	74.5	82.6	85.9	84.6	77.6	72.8	64.8	55.4
		MCWB	49.0	49.9	53.3	57.1	64.2	71.2	72.1	72.6	67.9	65.2	58.7	50.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.5	56.3	58.0	63.1	70.5	79.4	78.9	79.6	74.7	71.6	66.3	61.3
		MCDB	59.3	59.9	62.9	70.5	79.6	86.6	85.8	87.3	82.2	76.7	70.3	64.3
	2%	WB	54.5	54.3	56.5	61.2	68.6	76.4	77.0	77.8	72.8	69.7	63.2	55.8
		MCDB	57.4	58.0	61.1	67.3	77.4	84.6	84.4	85.4	79.2	74.3	67.6	58.8
	5%	WB	52.2	52.7	55.5	59.8	67.1	74.4	75.5	76.5	71.1	68.1	61.2	53.9
		MCDB	55.4	56.4	59.9	65.4	75.5	82.7	83.3	83.6	76.9	72.9	66.1	56.9
	10%	WB	49.9	50.8	54.5	58.4	65.5	72.5	74.3	75.2	69.6	66.5	59.4	51.6
		MCDB	53.6	54.7	59.1	63.8	72.9	81.2	82.8	81.8	75.6	71.6	64.0	55.1
Mean Daily Temperature Range	5% DB	MDBR	11.9	12.5	11.6	12.2	13.2	13.4	13.7	13.1	12.9	12.7	11.9	10.0
		MCDBR	14.0	14.0	14.2	15.8	17.5	17.4	18.5	16.3	15.8	14.5	12.4	11.8
		MCWBR	10.1	10.3	9.0	8.5	9.4	9.2	8.5	8.6	8.9	8.6	8.4	8.8
	5% WB	MCDBR	11.6	12.4	12.4	13.5	15.6	15.4	14.6	14.4	13.3	12.2	10.5	10.3
		MCWBR	9.2	9.9	9.0	8.4	9.2	9.4	8.9	9.8	9.3	8.6	8.3	8.4
Clear-Sky Solar Irradiance	taub	0.341	0.364	0.406	0.456	0.444	0.425	0.426	0.440	0.422	0.394	0.367	0.342	
	taud	2.445	2.368	2.255	2.132	2.238	2.338	2.325	2.278	2.324	2.378	2.403	2.442	
	Ebn at Noon	253	264	265	259	266	270	268	261	257	250	240	241	
	Edh at Noon	25	31	39	47	43	39	40	40	36	30	26	23	
All-Sky Solar Radiation	RadAvg	543	789	1184	1554	1939	2224	2289	2016	1457	1004	630	484	
	RadStd	57	117	155	156	144	132	104	112	108	95	60	83	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	-144	N/A	N/A

Nomenclature: See separate page